Jiatuo "Charlie" Zou

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EDUCATION

Columbia University — New York, NY
Master of Science in Computer Science — Sep 2024 – May 2026
GPA: 4.00.

University of Illinois at Urbana-Champaign — Champaign, IL
Bachelor of Science in Computer Science — Aug 2021 – May 2024
GPA: 3.91 (Dean's List & Graduated with Highest Honors in Engineering).

PUBLICATIONS

- Augmenting Human Performance with an XR Agent Learning from Online Behavior and BCI Evidence. Z. Li*, X. He*, H. Chen*, **C. Zou***, et al. ACM UIST 2026 (accepted).
- Delegate the Details: Flexible Assembly Constraint Authoring for Collaborative XR. B. Yang, X. He, **C. Zou**, J.-S. Liu, B. Tversky, S. Feiner. IEEE ISMAR 2026 (accepted).
- B. Yang, X. He, **C. Zou**, J.-S. Liu, B. Tversky and S. Feiner, "Virtual Reality Task Guidance Through Relative 6DoF Pose Specification," 2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct).
- Yang, B., He, X., **Zou, C.**, Liu, J., Tversky, B., Feiner, S. Bimanual XR Specification of Relative and Absolute Assembly Hierarchies for Teleoperation. arXiv:2603.01495, 2026.
- B. Yang, S. Liu, **C. Zou**, X. He, C. Elvezio, S. Feiner. Cascade: Spatial Routing Abstractions for Human Developers and AI Coding Agents. ACM CHI 2027 (submitted, under review).

RESEARCH EXPERIENCE

SPICE Lab — Northwestern University, Evanston, IL (On-site)
Research Intern — Jun 2026 – Present
Project: SensorClaw

- Building an **Android** app in **Kotlin** and **Jetpack Compose** that turns a one-sentence request ("Buzz me when I knock the table") into a personal smartwatch activity and gesture tracker, working end to end on **Pixel Watch 3** and **Pixel 9**.
- Designed a **multi-agent LLM** pipeline that splits the request into trigger, condition, and action, trains a personal **IMU** detector from user-recorded motions, retrieves relevant APIs, and critiques it on held-out trials, with self-adaptation through feedback loops; targeting UIST 2027.

Project: Spatial Memory

- Designing an agent-assisted handoff system for shared workspaces, where users leave notes anchored to objects, places, or projects (a 3D printer) and an **LLM agent** filters them by person, project, and task, tracks the latest reported state, and summarizes what the next user needs to know.

Laboratory for Intelligent Imaging and Neural Computing — Columbia University, NYC, NY (On-site)
Research Assistant — May 2025 – Present
Project: OLIVE

- Developed a **Unity VR** "Spaceshooter" game integrating real-time EEG + gaze for target identification assistance.
- Built **gRPC-based** streaming to transmit gaze points and EEG signals from Unity to backend for online inference and training.
- Designed and implemented a **CLIP**-driven AI agent (OLIVE) that learns player intent from **gaze-EEG** correlations, classifying targets into 2 categories – friendlies or enemies – using these neural feedbacks to provide dynamic visual cues.
- Conducted controlled user studies demonstrating rapid model convergence within ~80 seconds, achieving greater than 0.90 AUC in target classification and enabling personalized target suggestions in real time.

Computer Graphics and User Interface Lab — Columbia University, NYC, NY (On-site)
Research Assistant — Sep 2024 – Present
Project: Delegate the Details

- Constructed a **Unity VR** system for robotic teleoperation comparing absolute (direct manipulation) and relative (delegation-based) control modes for 6 DoF object placement and hierarchical manipulation in VR.
- Implemented bimanual interaction, object grouping, and part snapping assistance for complex 3D assembly tasks.
- Evaluated a user study (n=20), with relative mode reducing workload (NASA-TLX p=.003), achieving 85% user preference, and improving overall task efficiency by 32% through all difficulties.

Project: Cascade Messaging Toolkit

- Designed and implemented 3 example case scenarios (e.g., robot safety alerts, IoT monitoring, drone command tasks) to evaluate spatial and hierarchical messaging workflows in **Unity**.
- Compared Cascade vs. **Unity Events** by integrating Tobii eye tracking (120 Hz), timed coding tasks, and post-study surveys to evaluate correctness, workload, and usability across these 2 paradigms.
- Conducted and analyzed a within-subject study (n=14), showing Cascade improved task correctness (+0.34, $p < .001$) and reduced cognitive load while maintaining practical usability.

Lab of Brain-Computer Intelligence Hangzhou Dianzi University, China (On-site)
Research Assistant May 2024 – Aug 2024
Project: IDTS Healthcare "Pod"

- Built a full-stack system for an **Intelligent Diagnosis & Telemedicine System (IDTS)** for public health measuring, consisting of 3 components: an onboard **React** web UI, a WeChat mini-program for profile tracking, and a **MySQL** backend.
- Enabled real-time acquisition of 5 health data structures (ECG, temperature, BP, SpO_2, glucose) via **Bluetooth** Linktop SDK.
- Visualized multi-modal data using **Apache ECharts** (line charts, histograms) in mini-program.
- Led a team of 7 people on project and assigned tasks to everyone based on expertise; planned an 8-week milestone check for teammates and checked progress weekly during meetings, including a periodic 1-to-1 dev check helper session.

Forward Data Lab UIUC, Champaign, Illinois (On-site)
Research Assistant Aug 2023 – Dec 2023
Project: LLM Context Augmentation

- Built a **RAG**-based **LLM** pipeline using **LangChain + Chroma** vector store, embedding both document chunks (papers) and conversation history with OpenAI **text-embedding-ada-002**.
- Experimented with open-source models (e.g., **Llama-2-7b**) to test its summarization capability on self-designed questions.
- Deployed the pipeline onto the lab server and implemented metrics like **BLEU** to test the system against state-of-the-art private models like **ChatGPT-3**'s outputs as reference, achieving ~40 **BLEU score**.

WORK EXPERIENCE

COMS4731 Computer Vision I Columbia University, NYC, NY (On-site)
Teaching Assistant Sep 2025 – Dec 2025

- Supported the course by holding weekly office hours, grading biweekly homework, proctoring exams, and maintaining course materials, such as publishing homework and grades.

PhinD Champaign, Illinois (Hybrid)
Software Engineer Intern Jan 2025 – May 2025

- Built a file management system with **AWS S3** integration, implementing real-time file preview functionality.
- Managed **PostgreSQL** database on **AWS RDS**, implementing CRUD operations via **SQLAlchemy ORM**.
- Developed efficient **RESTful API** endpoints with **FastAPI** to optimize frontend-backend communication.
- Implemented a job recommendation system using text embeddings and cosine similarity, with additional keyword filtering.

CS361 Probability and Statistics UIUC, Champaign, Illinois (On-site)
Course Assistant Aug 2023 – May 2024

- Graded a total of 5 homework assignments, proctored midterm and final exams, and hosted regular office hours.
- Constructed and trained an encoder model in **PyTorch** to analyze and predict student grades based on past homework grades.

RELATED PROJECTS

EIT/IMU Hand-Pose Data Collection Northwestern University, Evanston, IL (On-site)
Research Project 2026

- Collected synchronized **EIT** and **IMU** wrist data with motion-capture hand-pose ground truth (**EasyMocap**) for wearable hand-pose estimation.

Dungeon Builder/Crawler Columbia University, NYC, NY (On-site)
Course Project Jan 2025 – May 2025

- Built a **mobile AR** dungeon-builder app using **ARCore**, supporting 6DoF object placement (floors, doors, enemies, chests) with hierarchical placement constraints (objects to be placed on a floor).
- Designed an adaptive instruction UI that tracks task progress and updates completion status dynamically.
- Implemented undo/redo, delete, and **JSON**-based save/load features, enabling levels to be imported directly into a **Meta Quest VR** dungeon-crawler app for an immersive game experience.

TrippinDark UIUC, Champaign, Illinois (On-site)
Course Project Jan 2023 – May 2023

- Implemented **AI** behavior for NPCs to support dynamic gameplay and player interaction in **Unreal Engine 5**.
- Designed and built two game levels using free assets, integrating sound effects and custom animations for immersive experience.
- Developed core gameplay features including inventory systems, interactive objects (pick-ups, puzzles), and trap mechanisms, enhancing game complexity and engagement for users.

AWARDS

- Awards: Winner of General Track at HackIllinois 2022, Web Development.

SKILLS

Languages: Python, C++, C#, Java, JavaScript, TypeScript, SQL, HTML, CSS, Kotlin, Swift.
Frameworks & Tools: Unity, Git, CI/CD, PostgreSQL Database, React.js, Express.js, Node.js, Firebase, GCP, FastAPI, AWS.
AI: Machine Learning, Deep Learning, Computer Vision, RL, NLP, Transformers, LLM, TensorFlow, PyTorch, Claude, Codex.